



CERTIFICATE OF CALIBRATION

FT-3001: Fuel gas header

Certificate number	GCS-CAL-25-0117
Customer	Dunes Energy Company, Southern Dunes Field Central Processing Facility 2 (CPF-2)
Instrument	Ultrasonic, 4-path, DN300
Tag / serial number	FT-3001 / US4-300-18842
Measuring range	0-18,000 Sm ³ /h
Date of calibration	18 January 2025
Next calibration due	31 January 2026
Method	In-situ verification per ISO 17089-1:2019: zero-flow verification, speed-of-sound check against value calculated from gas composition (AGA Report No. 10), path and transducer diagnostics, transmitter and flow computer checks
Reference standards	Pressure calibrator PC-118 (cert. GCS-REF-24-311), temperature reference TR-044 (cert. GCS-REF-24-298), traceable to SI via national standards

Results

Check	As found	As left	Acceptance limit	Result
Flow measurement error (% of reading)	+0.41%	+0.12%	±1.0%	PASS
Speed of sound deviation (average of paths)	0.03%	0.01%	±0.2%	PASS
Zero-flow reading	0.003 m/s	0.001 m/s	±0.006 m/s	PASS
Transmitter / flow computer check	Within limits	Within limits	Manufacturer spec.	PASS

The instrument was found to be within the acceptance limits after adjustment and is fit for use until the next calibration due date shown above. Results relate only to the item calibrated at the time of calibration.

Joseph Mathew

Joseph Mathew
Calibration Technician
18 January 2025

Ahmed Fawzy

Ahmed Fawzy
Technical Manager (approved)
18 January 2025





CERTIFICATE OF CALIBRATION

FT-5101: HP flare header

Certificate number	GCS-CAL-24-0562
Customer	Dunes Energy Company, Southern Dunes Field Central Processing Facility 2 (CPF-2)
Instrument	Ultrasonic flare gas meter, 2-path, DN600
Tag / serial number	FT-5101 / UFG-600-07731
Measuring range	0.03-100 m/s
Date of calibration	28 May 2024
Next calibration due	31 May 2025
Method	In-situ verification per ISO 17089-2:2012: zero-flow verification, speed-of-sound check against value calculated from gas composition (AGA Report No. 10), path and transducer diagnostics, transmitter and flow computer checks
Reference standards	Pressure calibrator PC-118 (cert. GCS-REF-24-311), temperature reference TR-044 (cert. GCS-REF-24-298), traceable to SI via national standards

Results

Check	As found	As left	Acceptance limit	Result
Flow measurement error (% of reading)	+1.90%	+0.60%	±3.0%	PASS
Speed of sound deviation (average of paths)	0.15%	0.05%	±0.2%	PASS
Zero-flow reading	0.003 m/s	0.001 m/s	±0.006 m/s	PASS
Transmitter / flow computer check	Within limits	Within limits	Manufacturer spec.	PASS

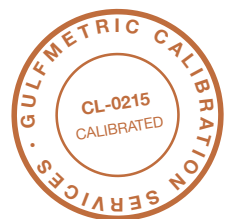
The instrument was found to be within the acceptance limits after adjustment and is fit for use until the next calibration due date shown above. Results relate only to the item calibrated at the time of calibration.

Joseph Mathew

Joseph Mathew
Calibration Technician
28 May 2024

Ahmed Fawzy

Ahmed Fawzy
Technical Manager (approved)
28 May 2024





CERTIFICATE OF CALIBRATION

FT-5102: LP flare header

Certificate number	GCS-CAL-25-0188
Customer	Dunes Energy Company, Southern Dunes Field Central Processing Facility 2 (CPF-2)
Instrument	Ultrasonic flare gas meter, 2-path, DN300
Tag / serial number	FT-5102 / UFG-300-09215
Measuring range	0.03-100 m/s
Date of calibration	10 February 2025
Next calibration due	28 February 2026
Method	In-situ verification per ISO 17089-2:2012: zero-flow verification, speed-of-sound check against value calculated from gas composition (AGA Report No. 10), path and transducer diagnostics, transmitter and flow computer checks
Reference standards	Pressure calibrator PC-118 (cert. GCS-REF-24-311), temperature reference TR-044 (cert. GCS-REF-24-298), traceable to SI via national standards

Results

Check	As found	As left	Acceptance limit	Result
Flow measurement error (% of reading)	+1.20%	+0.50%	±3.0%	PASS
Speed of sound deviation (average of paths)	0.10%	0.04%	±0.2%	PASS
Zero-flow reading	0.003 m/s	0.001 m/s	±0.006 m/s	PASS
Transmitter / flow computer check	Within limits	Within limits	Manufacturer spec.	PASS

The instrument was found to be within the acceptance limits after adjustment and is fit for use until the next calibration due date shown above. Results relate only to the item calibrated at the time of calibration.

Joseph Mathew

Joseph Mathew
Calibration Technician
10 February 2025

Ahmed Fawzy

Ahmed Fawzy
Technical Manager (approved)
10 February 2025

